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Subject: Data Structures and Algorithms (CS201) Internal Assessment 1

Ans 1(a).

Stack is a data structure that works on LIFO order, the last element inserted is removed first. It is used in undo operation of an editor and also to check balanced brackets.

Ans 1(b).

Stack - LIFO, queue - FIFO. In stack we push and pop from the top. In queue elements are added at rear and removed from front. Stack example is books piled up, queue example is a queue at a bank.

Ans 3(a).

BFS: start from source, put it in a queue. Take a node out from the queue, and add all its neighbours which are not visited to the queue. Repeat till queue becomes empty.

Ans 3(b)

BFS uses queue and DFS uses stack. BFS goes level wise and DFS goes deep first.

Ans 2

Binary search is a searching method in which we find the middle of the array and compare the key with it. If the key is smaller we go to the left part otherwise the right part. This is repeated. Because we keep dividing the array into two parts the complexity is $O(\log n)$.

Ans 4.

For reversing a linked list we change the direction of the pointers so that each node points to the previous node instead of the next node. We take pointers prev and curr and move through the list. The time taken is $O(n)$.

